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Occurrence of black corals (order: Antipatharian) in Andaman and Nicobar Island, India

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A total of 15 species of antipatharians were accounted from the study sites. Maximum number of species was reported in Henry Lawrence Island and minimum in Paget Island, Rowe Island and North Reef Island. The present study yielded seven species (*Cupressopathes abies*, *C. pumila*, *Tanacetipathes hirta*, *Antipathes dendrochristos*, *Cladopathes plumose*, *Trissopathes tetracrada*, *Dendrobathypathes boutillieri*) and all are first report to India.

[Key words: Hexacoral, Antipatharians, Black corals, Andaman and Nicobar]

Introduction

The Antipatharians are commonly called as black corals or horny corals look like an upright and bushy growth forms and mimic gorgonian octocorals. Most of growth forms inhabit in the shallow tropical deep sea environment, their ecology and distribution are largely unknown¹⁻¹⁰. Black corals are suspension plankton feeders and mostly present in shady or dark habitats, attached over the hard substrate¹¹⁻¹⁴ and the relationship with abiotic environmental features¹⁵. In Caribbean Sea, black corals reported below the euphotic zone¹⁶ and most of the species reported in shallow reef environment along the Indo – Pacific coast¹⁷⁻²².

The taxonomic studies on black corals along the seacoast of Indonesian Archipelago can be found in two detailed monographs²³⁻²⁴ and also reported in the Northern Moluccas²⁵, in Northeast Atlantic²⁶, Bay of Biscay²⁷ and Maldives and Lakshadweep Archipelagos²⁸⁻²⁹. Shallow water black corals reported from Indo-Malayan region²². A total of eight species belonging two families and seven genera were recorded from Andaman and Nicobar Islands³⁰.

Black corals are very numerous in shallow tropical reef communities and are well documented in deep benthic communities, where the colonies may provide a substrate in habitats with poor complexity. Present study

is to assess the community structure of antipatharian and reporting 15 species belonging to 11 genera, four families from Andaman and Nicobar Island.

Material and Methods

Andaman and Nicobar Archipelago have 572 reef islands between Myanmar (North) to Sumatra (South) 6° to 14° North and 92° to 94° East with 2000 km² coral reef cover (Figure 1). The black coral taxonomical study was conducted in different reef islands and samples were collected during June 2013 to July 2016 at a depth range of 5.0 – 40.0 m with the help of SCUBA. The specimens were identified based on the morphological characteristics such as growth form, colour, shape, size, colour of the polyp, diameter of the axis and spines arrangement¹⁶. Live colonies photo made by underwater digital camera (Canon G10) and spine arrangement of each species was examined under Leica –DFC 500 stereo microscope. The identified samples were deposited in the National Zoological Collection of ZSI, ANRC, Port Blair. Quantitative data were obtained at each study sites by using the adapted belt transect method and counting all the colonies observed along 5 transects with 2.0 m wide³¹ (Including 1.0 m on each side of the diver) at depths of 5.0 to 40.0 m.

Results

A total of 15 species of antipatharians belonging 11 genera and four families was reported from 11 study sites (Figure 1- 5, Table 1) and a maximum number (6) of species, was recorded from Henry Lawrence Island in South Andaman followed by Peacock Island (4), Trilbay Island (4), Havelock Island (4), Shark Island (3), Oliver Island (3), Sound Island (3), Landfall Island (2), Paget Island (1), Rowe Island (1), North Reef Island (1). The species *Cirrhipathes anguina* was common in most of the Islands studied (Table 1). Percentage of cover was maximum (12.5 %) in Peacock Island, North Andaman followed by Shark Island (10 %), Henry Lawrence Island (10 %), North Reef Island (7.5 %), Trilbay Island (7.5 %), Sound Island (7.5 %), Havelock Island (7.5 %), Paget Island (6.25 %), Landfall Island (6.25 %), Oliver Island (5 %) and Rowe Island (3.75 %) (Figure 6).

SYSTEMATIC POSITION

Phylum: Cnidaria Hatschek, 1888
 Class: Anthozoa
 Sub-Class: Hexacorallia
 Order: Antipatharia
 Family: Myriopathidae Opresko, 2001
 Genus: *Antipathella* Brook, 1889

1. *Antipathella subpinnata* (Ellis and Solander, 1786) (Fig 2 A, B)

1786 *Antipathes subpinnata* Ellis & Solander

Material Examined: ZSI/ANRC 14295 (Henry Lawrence, South Andaman Island, Lat.12°05.000 N; Long. 93°06.312 E), sample 18.5 cm height, 13.0 cm width, dried preservation ZSI/ANRC 5612 (Long Island, Lat. 12°21.749 N; Long. 92°55.410 E)

Description: see Ellis and Solander,³²; Tocino *et al.*,³³. The colony, brownish yellow colour in under water, densely branched, small branches not even in size, the longest branches 13.0 cm long, the spines are needle like, 0.12 mm in size, polyps are arranged uniserially and 7 to 9 polyps per centimeter along the axis.

Distribution: France, Mediterranean Sea, Mediterranean Sea – Eastern Basin, New Zealand, North Atlantic Ocean, Spain, India : Andaman and Nicobar.

Genus: *Cupressopathes*

2. *Cupressopathes abies* (Linnaeus, 1758) (Fig 2 C, D)

1758 *Antipathes abies* Linnaeus

1758 *Gorgonia abies* Linnaeus

Material Examined: ZSI/ANRC 13030 (Landfall Island, North Andaman, Lat. 13°38.501 N; Long. 92°59.812 E), the colony 24.0 cm long, 7.0 cm breadth, dry preserved sample, 10.0 m depth.

Description: See Linnaeus,³⁴. The colony is bottlebrush pinnulation and sparsely branched, more than 60.0 cm height, pinnular and subpinnular not uniform, up to 7.0 cm long, 1.0 - 1.6 mm in diameter at base, 5 -7 primary pinnules per centimeter, polyps are single series, 0.8 – 0.9 mm in transverse diameter, 10 – 12 per centimeter.

Table 1 — Distribution of Antipatharian corals in surveyed areas of Andaman and Nicobar Islands

S.No	Name of Species	1	2	3	4	5	6	7	8	9	10	11
1	<i>Antipathella subpinnata</i>										+	
2	<i>Cupressopathes abies</i>			+								
3	<i>Cupressopathes gracilis</i>				+							
4	<i>Cupressopathes pumila</i>					+						
5	<i>Myriopathes antrocrada</i>					+					+	+
6	<i>Plumapathes pennacea</i>							+	+	+	+	
7	<i>Tanacetipathes hirta</i>								+		+	
8	<i>Antipathes dendrochristos</i>									+		
9	<i>Antipathes elegans</i>										+	
10	<i>Cirrhipathes anguina</i>	+	+	+	+	+	+	+	+	+	+	+
11	<i>Cirrhipathes contorta</i>				+							+
12	<i>Stichopathes solorensis</i>											+
13	<i>Cladopathes plumose</i>							+				
14	<i>Trissopathes tetracrada</i>							+				
15	<i>Dendrobathypathes isocrada</i>					+						
Total		1	1	2	3	4	1	4	3	3	6	4

(1 – Paget Island, 2 – Rowe Island, 3 – Landfall Island, 4 – Shark Island, 5 – Peacock Island, 6 – North reef Island, 7 – Trilbay Island, 8 – Oliver Island, 9 – Sound Island, 10 – Henry Lawrence, 11 – Havelock Island)

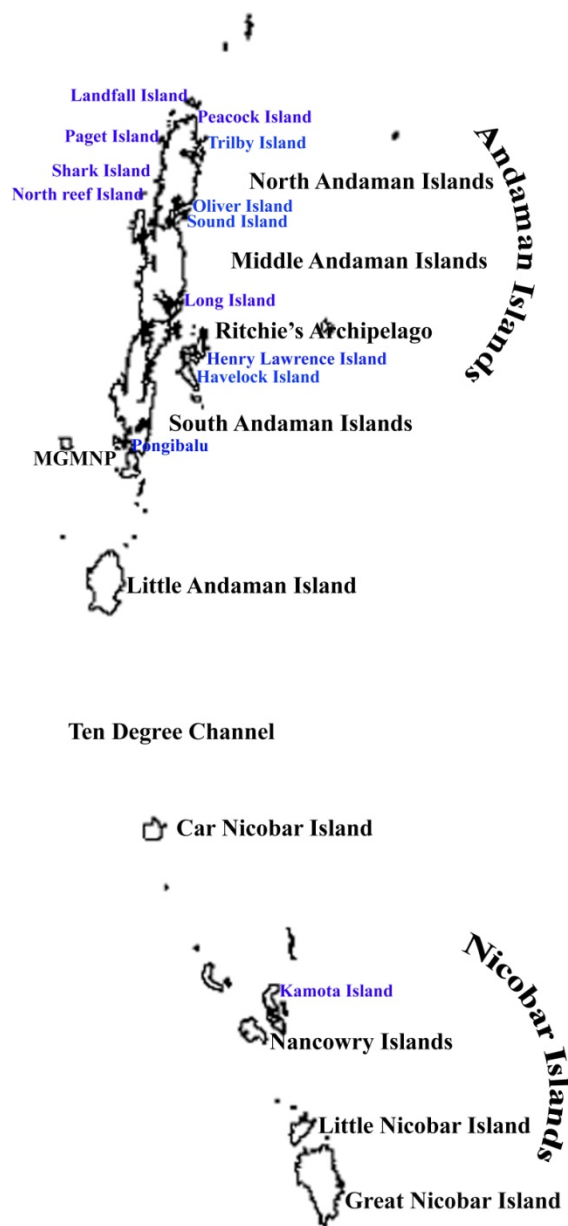


Fig. 1 — Location – study area

Distribution: Indo-Pacific, Western Pacific, Indian Ocean, New Zealand, India: Andaman and Nicobar.

3. *Cupressopathes gracilis* (Thomson and Simpson, 1905) (Fig 2 E, F)

1860 *Antipathes gracilis* Gray

Material Examined: ZSI/ANRC 14303 (Shark Island – North Andaman, Lat. 13°12.064 N Long. 92°45.255 E) the dry preserved sample 28 cm height, 11 cm breadth, 12 m depth; ZSI/ANRC: 5606 (Pongibalu at Port Blair Lat. 11°30.956 N; Long. 92°39.201 E)

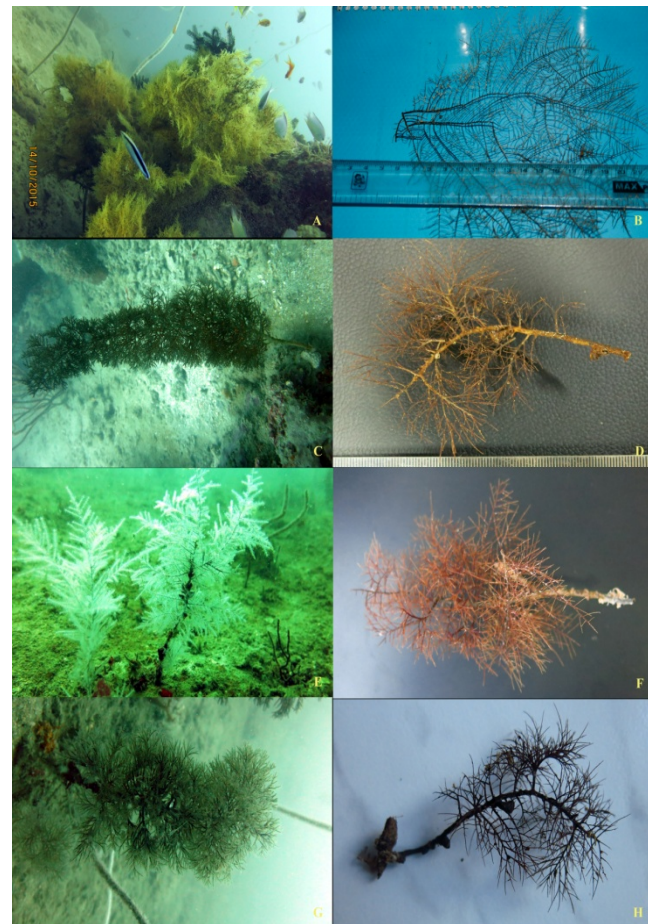


Figure 2. Antipatharian corals live and preserved samples, A & B - *Antipathella subpinnata*, C & D - *Cupressopathes abies*, E & F - *Cupressopathes gracilis*, G & H - *Cupressopathes pumila*

Description: see Thomson and Simpson,³⁵. Living colonies orange or yellowish brown colour and branched colonies, height 1.5 m and 60.0 cm wide, inter node of branches 4.29 to 5.26 mm, spines small, smooth, mostly 0.1 – 0.15 mm in size, polyp are very small tentacles.

Distribution: Gulf of Mexico, Indian Ocean, New Zealand, North Atlantic Ocean, Bermuda, India: Andaman and Nicobar.

4. *Cupressopathes pumila* (Brook, 1889) (Fig 2 G, H)

1889 *Antipathes pumila* Brook, 1889

Material Examined: ZSI/ANRC: 14657 (Peacock Island – North Andaman, Lat. 13°33.951 N; Long. 93°03.153 E), size of the sample 22.0 cm height, 8.0 cm breadth, dry preserved, 15.0 m depth.

Description: see Brook²³. The colony differs from *C. abies*, *C. gracilis*, 40.0 – 60.0 cm height in live, the branches arise from the main axis, polyps are white

colour, primary pinnules developing into side branches. The size of the colony is smaller than other species, interpolyp space 0.2 mm or less, tentacles up to 0.7 mm.

Distribution: West Pacific, India: Andaman and Nicobar.

Genus: *Myriopathes*

5. *Myriopathes antrocrada* (Opresko, 1999) (Fig 3 A, B)

1999 *Antipathes antrocrada* Opresko

Material Examined: ZSI/ANRC 14297, Henry Lawrence, South Andaman Island, (Lat.12°05.000 N; Long. 93°06.312 E), the preserved sample 45.0 cm tall and 30.0 cm wide, 25.0 m depth; ZSI/ANRC 14659 Peacock Island, North Andaman, Lat. 13°33.951 N; Long. 93°03.153 E), 32.0 cm height and 18.0 cm width, 15.0 m depth; ZSI/ANRC 5610 (Havelock Island, Lat. 12°03.334 N; Long. 92°57.716 E).

Description: see Opresko³⁶. Live colonies more than 1.0 meter height, reddish brown colour, the basal stem diameter 3.3 mm, large branches up to 9.0 cm long, spines conical and horn shaped, 0.14 mm to 0.18 mm size, polyps slightly elongated in the transverse axis, tentacles knob like and 0.15 mm long.

Distribution: Indonesia, Australia, India: Andaman and Nicobar Islands

Genus: *Plumapathes*

6. *Plumapathes pennacea* (Pallas, 1766) (Fig 3 C, D)

1766 *Antipathes pennacea* Pallas

Material Examined: ZSI/ANRC 14296, Henry Lawrence, South Andaman Island, (Lat.12°05.000 N; Long. 93°06.312 E), 14cm height, 11.0 cm width dry preserved sample, 15.0 m depth; ZSI/ANRC 13029, Trilbay Island, North Andaman, (Lat. 13°24.675 N; Long. 93°04.245 E), 18.0 cm long and 12.0 cm breadth sample, 18.0 m depth; ZSI/ANRC 14646, Oliver Island (Lat. 13°00.038 N; Long. 92°59.216 E), 32.0 cm height and 21.0 cm breadth, 22.0 m depth; ZSI/ANRC 14655, Sound Island (Lat. 12°56.263 N; Long. 92°57.049 E), 15.0 cm long and 12.0 cm width sample, 8.0 m depth; ZSI/ANRC 5611, Long Island (Lat. 12°21.749 N; Long. 92°55.410 E).

Description: see Pallas³⁷. Live colonies brownish red colour in under water and densely branched, more than one meter height, primary pinnules simple and uniform in size, the spines are needle like in surface of branches and stems, polyps are arranged in a single series, 0.7 to 0.8 mm in size.

Distribution: Indo-Pacific, Caribbean Sea, Gulf of Mexico, New Zealand, South Atlantic Ocean, India: Andaman and Nicobar.

Genus: *Tanacetipathes*

7. *Tanacetipathes hirta* (Gray, 1857) (Fig 3 E, F)
1857 *Antipathe hirta* Gray, 1857

Material Examined: ZSI/ANRC 14645, Oliver Island (Lat. 13°00.038 N; Long. 92°59.216 E), the sample 22.0 cm height and dried sample, 22.0 m depth.

Description: Gray,³⁸. live colony more than 1.0 meter in size, uneven sparsely branched, main axis brown in colour and polyps are white colour in underwater, average size of axis diameter is 3.0 mm, spines are smooth, conical, 0.12 – 0.3 mm, polyps 0.22 – 0.51 mm long and 10-16 polyps in per centimeter, tentacles 0.12 – 0.36 mm long, mouth usually sagittally elongated.

Distribution: Gulf of Mexico, Bermuda, West Central Atlantic, West Indies, Brazil, India: Andaman and Nicobar

Family: Antipathidae Ehrenberg, 1834



Figure 3. — A & B - *Myriopathes antrocrada*, C & D - *Plumapathes pennacea*, E & F - *Tanacetipathes hirta*, G & H - *Antipathes dendrochristos*.

Genus: *Antipathes* Pallas, 1766

8. *Antipathes dendrochristos* Opresko, 2005 (Fig 3 G, H)

Material Examined: ZSI/ANRC: 14656, Sound Island, North Andaman (Lat. 12°56.263 N; Long. 92°57.049 E), 18.0 cm height and 9.0 cm width sample, dried preserve sample, 8.0 m depth.

Description: see Opresko³⁹. live colony bushy with multibranched, more than 60.0 cm height, axis reddish brown in colour, polyp white colour, spines small, conical, smooth, 0.07 – 0.09 mm in size, polyps 1.0 – 1.4 mm in transverse diameter, short, blunt tentacles.

Distribution: North Pacific Ocean, United States, India: Andaman and Nicobar.

9. *Antipathes elegans* (Thomson and Simpson, 1905) (Fig 4 A, B)

1905 *Antipathella elegans* Thomson and Simpson

Material Examined: ZSI/ANRC: 14294, Henry Lawrence, South Andaman (Lat. 12°05.000 N; Long. 93°06.312 E), 42.0 cm height, dried preserve sample, 15.0 m depth; ZSI/ANRC 5613 (Kamota Island Lat. 08°02.183 N; Long. 93°32.573 E)

Description: see Thomson and Simpson,³⁵. Living colonies are orange colour, more than 1.5 m height and 1.0 m breadth, multi branched, bushy colonies reaching a size of 1.0 m or more and branches free, spines are small, conical, smooth, mostly 0.08 – 0.1 mm in size, polyps 1.0 - 1.4 mm in transverse diameter with short tentacles.

Distribution: Indian Ocean, India: Andaman and Nicobar Islands.

Genus: *Cirripathes* de Blainville, 1830

10. *Cirripathes anguina* (Dana, 1846) (Fig 4 C, D, Fig. 5 H)

1846 *Antipathes anguina* Dana

Material Examined: ZSI/ANRC: 14293, Henry Lawrence (Lat. 12°05.000 N; Long. 93°06.312 E), 52.0 cm length sample, 15.0 m depth; ZSI/ANRC 14299, Rowe Island (Lat. 13°14.913 N; Long. 92°50.419 E), 34.0 cm long sample; ZSI/ANRC 14300, North reef Island (Lat. 13°05.785 N; Long. 92°43.355 E), 15.0 cm height sample, 12.0 m depth; ZSI/ANRC 14301, Shark Island (Lat. 13°12.064 N; Long. 92°45.255 E), 25.0 cm height sample; ZSI/ANRC 14304, Paget Island (Lat. 13°25.805 N; Long. 92°50.728 E), 12.0 cm height sample; ZSI/ANRC 13031, Peacock Island, North Andaman

(Lat. 13°33.951 N; Long. 93°03.153 E), 1m long sample; ZSI/ANRC 5607, Havelock Island (Lat. 12°03.334 N; Long. 92°57.716 E).

Description: see Kumar, *et al.*,³⁰. Live colonies unbranched and greenish brown colour, 60.0 – 100.0 cm in height, average axis size 3.9 mm, average spine size 0.09 mm, polyps pale brown or white colour, transverse diameter about 1.0 – 2.0 mm arranged in multiple irregular rows, sagittal tentacles about 2.0 – 3.0 mm, lateral tentacles 1.05 – 1.4 mm and oral cone about 0.5 – 0.7 mm in size.

Distribution: Fiji Islands, Red Sea, Indo - Pacific, India: Andaman and Nicobar.

11. *Cirripathes contorta* van Pesch, 1910 (Fig 4 E – F, Fig. 5 G)

1910 *Cirripathes* (*Eucirripathes*) *contorta* Van Pesch

1910 *Cirripathes contorta* van Pech, 1910

Material Examined: ZSI/ANRC: 14302 (Shark Island), 32.0 cm height sample; ZSI/ANRC 5608

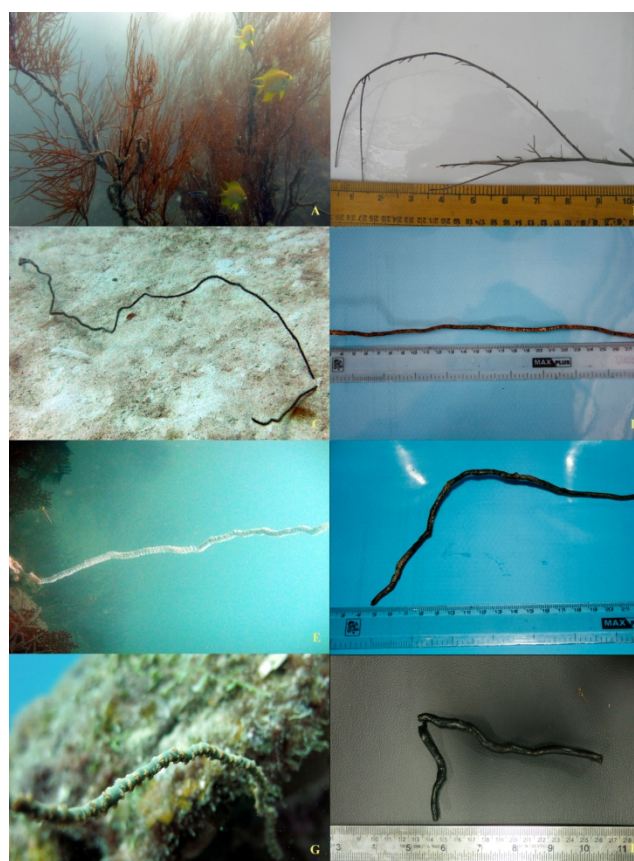


Figure 4. — A & B - *Antipathes elegans*, C & D - *Cirripathes anguina*, E & F - *Cirripathes contorta*, G & H - *Stichopathes solorensis*

(Pongibalu Lat. 11°30.956 N; Long. 92°39.201E), ZSI/ANRC 14647 (Havelock Island, South Andaman), 34.0 cm height sample dried preserve samples.

Description: Kumar, *et al.*,³⁰. Live colonies unbranched and white colour, 150.0 cm in height, average size 6.88 mm, spines 0.19 mm, polyps white colour and arranged in multiple irregular rows, sagittal tentacles about 1.3 – 2.15 mm, lateral tentacles 1.0 – 1.45 mm and oral cone about 0.4 – 0.6 mm in size.

Distribution: Banda Islands, Indo-Pacific, India: Andaman and Nicobar.

Genus: *Stichopathes* Brook, 1889

12. *Stichopathes solorensis* van Pesch, 1914 (Fig 4 G, H)

Material Examined: ZSI/ANRC: 5609 (Kamota Island Lat. 08°02.183N; Long. 93°32.573E), ZSI/ANRC 14648 (Havelock Island, South Andaman), 21.0 cm height dried preserve sample.

Description: see van Pesch,²⁴. Colonies unbranched and greenish grey colour, 1.8 m in height in live, axis

size 8.24 mm, spiral curving is irregular, spiral's height 9.0 - 57.0 cm, spines 0.13 mm in size, polyps pale orange in colour and arranged in multiple irregular rows, tentacles about 1.3 – 2.15 mm, and oral cone about 0.5 – 0.6 mm in size.

Distribution: Indo-Pacific, India: Andaman and Nicobar.

Family: Cladopathidae Kinoshita, 1910

Genus: *Cladopathes* Brook, 1889

13. *Cladopathes plumose* Brook, 1889 (Fig 5 A, B)

Material Examined: ZSI/ANRC 13027 (Trilbay Island, North Andaman, Lat. 13°24.675 N; Long. 93°04.245 E), 28.0 cm long and 6.0 cm breadth, dried sample.

Description: see Brook,²³. Live colony axis brown colour and polyps are white in colour, shrub like, stem and branches pinnulate and subpinnulate, 0.5 – 1.0 cm long pinnules, 16 pinnules per centimeter, 0.3 mm diameter pinnules, spines simple, smooth, acute, 0.06 – 0.08 mm in size, polyps 3.0 – 4.0 mm in transverse with three polyps per centimeters.

Distribution: Off Prince Edward Island, Off Ascension Island, Great Australian Bight, India: Andaman and Nicobar Islands.

Genus: *Trissopathes* Opresko, 2003

14. *Trissopathes tetracrada* Opresko, 2003 (Fig 5 C, D)

Material Examined: ZSI/ANRC 13028 (Trilbay Island, North Andaman, Lat. 13°24.675 N; Long. 93°04.245 E), 15.0 cm height and 8 cm width sample.

Description: see Opresko,²¹. Live colonies bushy, overlapping branches, axis and branches are pinnulate and subpinnulate, 0.5 – 1.0 cm long and 0.15 - 0.2 mm in diameter, four per centimeter, spines 0.08 - 0.10 mm height, three to four spines per millimeter, polyps are transversely elongated, 2.0 – 3.0 mm in diameter.

Distribution: New Zealand, South Australia, Tasmania, Eastern Atlantic off cape Verde, India: Andaman and Nicobar.

Family: Schizopathidae Brook, 1889

Genus: *Dendrobathypathes* Opresko, 2002

15. *Dendrobathypathes isocrada* Opresko, 2002 (Fig 5 E, F)

Material Examined: ZSI/ANRC 14658 (Peacock Island, North Andaman, Lat. 13°33.951 N; Long. 93°03.153 E), 8.0 cm height and width, dried preserve sample.

Description: see Opresko,²⁰. Live colonies white in colour, net branched, stem not distinct or slightly

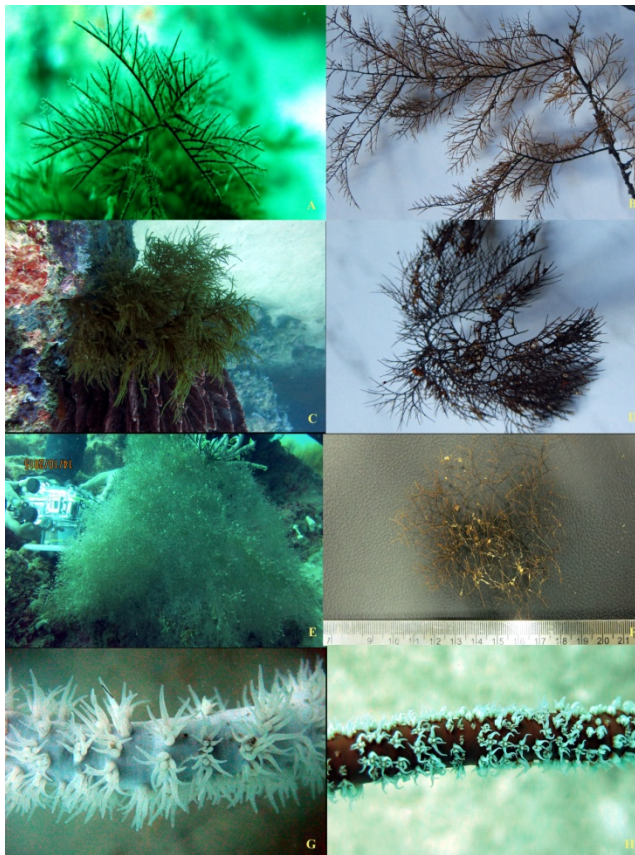


Figure 5. A & B — *Cladopathes plumose*, C & D - *Trissopathes tetracrada*, E & F - *Dendrobathypathes isocrada*, G - live polyp of *Cirrhipathes contorta*, H - Live polyp of *Cirrhipathes anguina*

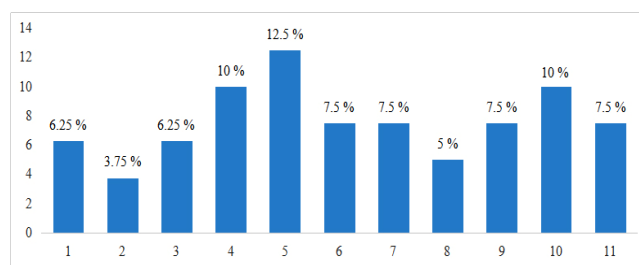


Figure 6. — Percentage of Antipatharian coral coverage from different study sites in Andaman and Nicobar Islands. (1 – Paget Island, 2 – Rowe Island, 3 – Landfall Island, 4 – Shark Island, 5 – Peacock Island, 6 – North reef Island, 7 – Trilbay Island, 8 – Oliver Island, 9 – Sound Island, 10 – Henry Lawrence, 11 – Havelock Island)

thicker than other branches, pinnules overlapping and adhering, spines simple, conical, compressed, 0.03 – 0.1 mm in size, polyps 2.5 – 4.0 mm in transverse diameter, two to three polyps per centimeter.

Distribution: Australia, New Zealand, India: Andaman and Nicobar.

Discussion

The Andaman and Nicobar Islands are rich and a variety of marine fauna and flora including endangered, schedule faunas. Two Marine National Parks, namely Mahatma Gandhi Marine National Park and Rani Jhansi Marine National Park; 96 wildlife sanctuaries have been established for protecting these ecosystems. Among them seven study sites coming under the Wildlife sanctuaries (Pagat Island, Shark Island, North Reef Island, Rowe Island, Peacock Island, Trilbay Island, Oliver Island) and Henry Lawrence Island comes under Rani Jhansi Marine National Park. Which have rich biodiversity sources of fauna and flora, including corals, soft corals, sponges, mollusc, echinodermata, Pisces, sea snakes, sea turtles, birds, marine mammals and so on.

Antipatharian are found in a wide range of habitats from the shallow to deeper water and commonly found in the reef areas of different depth. Distribution of black corals were based on light attenuation, substratum inclination, suspended food, flow regimes, wave exposure and water current. Very few antipatharians were reported from Andaman and Nicobar as well as in Indian reef regions. Present study reported the occurrence of antipatharians coral species along with their percentage cover in Andaman and Nicobar Islands.

A total 15 species belongs 3 families and 11 genera were reported from 11 study sites, of which seven species (*Cupressopathes abies*, *C. pumila*,

Tanacetipathes hirta, *Antipathes dendrochristos*, *Cladopathes plumose*, *Trissopathes tetracrada*, *Dendrobathypathes isocrada*) are new distribution report to the Indian coast and remaining eight species (*Antipathella subpinnata*, *Cupressopathes gracilis*, *Myriopathes antrocrada*, *Plumapathes pennacea*, *Antipathes elegans*, *Cirrhipathes anguina*, *C. contorta*, *Stichopathes solorensis*) were identified and reported by Kumar *et al.*,³⁰. The abundance of antipatharians cover is in the following order: Peacock Island > Shark Island > Henry Lawrence Island > North Reef Island > Trilbay Island > Sound Island > Havelock Island > Paget Island > Landfall Island > Oliver Island > Rowe Island.

The antipatharians are reported from Pearl Oyster bed of Gulf of Mannar³⁵ and associated shrimp from Andaman and Nicobar Islands⁴⁰. Extensive surveys are required to inventorise the antipatharians fauna to enhance the database of Andaman and Nicobar as well as India. The antipatharians occur on the steep slope and on inclined and shaded calcareous substrates. More studies are needed to signify their ecological role as a key parameter towards the understanding of the reef environment.

Conclusion

Antipatharian black corals are distributed worldwide at ocean depths varying from shallow to 360 m and they create critical habitat for invertebrates, fishes, including commensal species that are dependent upon black coral for their survival. The present study reported 15 species of antipatharians in shallow reef regions in Andaman and Nicobar Islands, of which, seven species are reported first time in Andaman and Nicobar as well as in India. The community structure of antipatharian was evaluated using standard ecological parameters. The widespread studies on antipatharian may signify their ecological role as a key parameter towards the understanding of the reef environment.

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